$$\int d^3p \; |p\rangle\langle p| = \mathbb{1}$$

$$S^\dagger = S = \lim_{T\to\infty} U(T,-T)$$

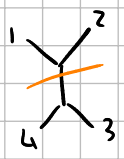
$$S = \mathbb{1} + iT$$

$$S^\dagger S = \mathbb{1} \rightarrow \mathbb{1} + iT^\dagger - iT + T^\dagger T = \mathbb{1}$$

$$\langle 0 | T | p_1 p_2 \cdots p_n \rangle = \mathcal{A}_n$$

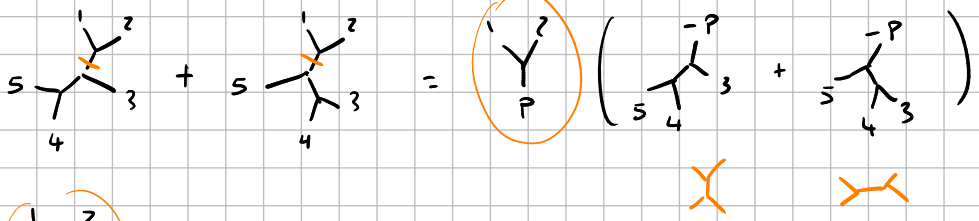
A_4 :

$\mathrm{Res}\, A_4 \,_{s_{12}=0} = s_{12}$

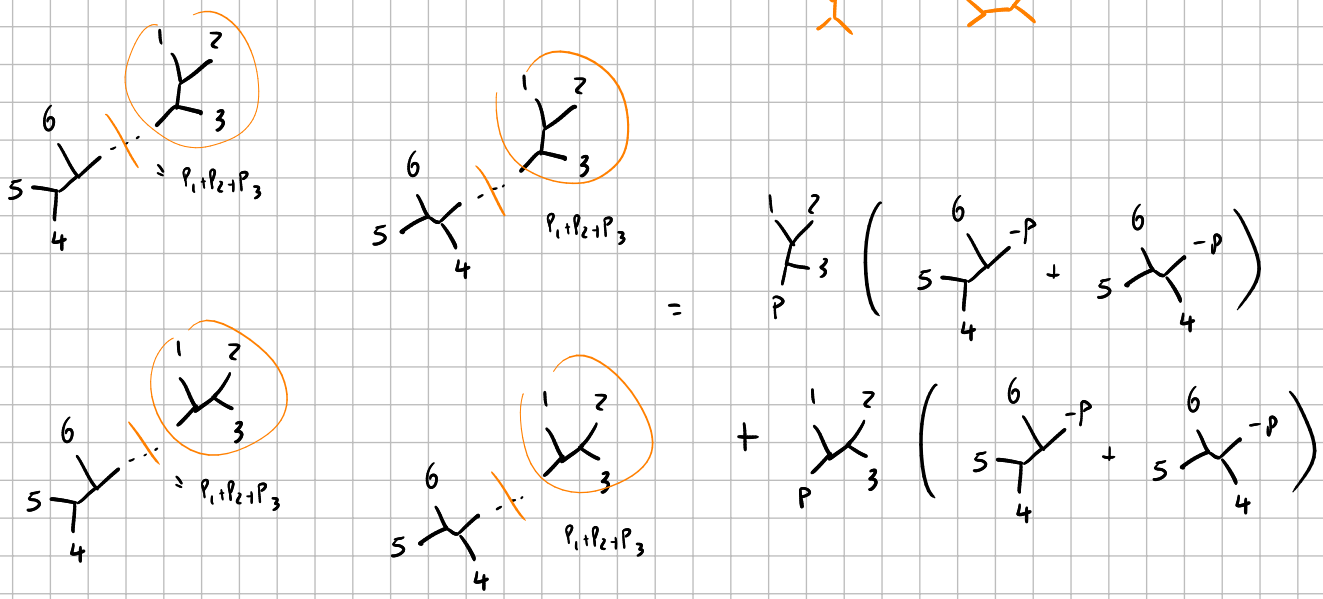


$= 1$

$\mathrm{Res}\, A_5 \,_{(s_{12}=0)} =$



$\mathrm{Res}\, A_6 (s_{123}=0)$



BCFW shift:

pick p_i, p_j among the external momenta $(i \neq j)$

send $p_i \to p_i + zq$ momentum

$p_j \to p_j - zq$

q s.t. $q^2 = 0$

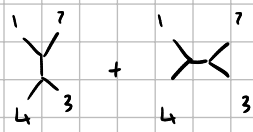
$q \cdot p_i = q \cdot p_j = 0$

ex: 4 particles momenta: $p_1 \quad p_2 \quad p_3 \quad p_4$

after shift: $p_1 + zq \quad p_2 - zq \quad p_3 \quad p_4$

- total momentum is still zero
- $(p_1 + zq)^2 = (p_2 - zq)^2 = 0$

amplitudes now depend on z

$A_4 =$ 

$= \frac{1}{s_{12}} + \frac{1}{s_{14}} = \frac{1}{p_1 \cdot p_2} + \frac{1}{p_1 \cdot p_4}$

after shift: $A_4(z) = \frac{1}{(p_1+zq)(p_2-zq)} + \frac{1}{(p_1+zq)\cdot p_4} = \frac{1}{p_1 \cdot p_2} + \frac{1}{p_1 \cdot p_4 + z\, p_4 \cdot q}$

$\Rightarrow = p_1 \cdot p_2$

look at $\frac{A(z)}{z}$ pole at $z=0$

$$\boxed{A(0) = 2\pi i \oint_{z=0} \frac{A(z)}{z}\, dz}$$

shift bap around $z=0$

$$\boxed{X_{1\,j+1}} = s_{12\cdots j} = (p_1 + p_2 + \cdots + p_j)^2 = (p_1+p_2)^2 + (p_3+p_4+\cdots)^2 + 2(p_1+p_2)\cdot(p_3+p_4+\cdots)$$

$$= s_{12} + s_{3\cdots j} + 2p_1\cdot(p_3+p_4\cdots) + 2p_2\cdot(p_3+p_4+\cdots)$$

$$= s_{12} + 2\underbrace{\sum_{i=3}^{j} p_1\cdot p_i}_{=s_{1j}} + 2\underbrace{\sum_{i=3}^{j} p_2\cdot p_i}_{2s_{2j}} + s_{3\cdots j}$$

$$= \boxed{s_{12} + \sum_{i=3}^{j} s_{1i} + \sum_{i=3}^{j} s_{2i} + s_{3\cdots j}}$$

$$\boxed{X_{2,i+1}} = (P_2 + \dots + P_i)^2 = (P_3 + P_4 + \dots)^2 + 2\, P_2 \cdot (P_3 + P_4 + \dots)$$

$$= \boxed{\sum_{i=3}^{i} s_{2i} + s_{3\dots i}}$$

$$\boxed{X_{2n}} = s_{23\dots n-1} = (P_2 + P_3 + \dots P_{n-1})^2 = (P_1 + P_n)^2 = 2 P_1 \cdot P_n$$

$$= 2 P_1 \cdot (-P_1 - P_2 \dots - P_{n-1})$$

$$= -2 P_1 \cdot P_2 - 2 P_1 (P_3 + P_4 + \dots P_{n-1})$$

$$= \boxed{-s_{12} - \sum_{i=3}^{n-1} s_{1i}}$$

(A) set $s_{1i} = 0$ $(i = 3, \dots, n-1)$

(B) $P_2 \to P_2 + q\, z$

$P_n \to P_n - q\, z$

(A_5) X_{13} X_{14} X_{24} X_{25} X_{35} $\left(\begin{matrix} X_{ii} = 0 & X_{i,i+1} = 0 \\ X_{ij} = X_{ji} & \end{matrix} \right)$

<u>shift:</u> $P_2 \to P_2 + q\, z$ $P_5 \to P_5 - q\, z$

effected if X_{ij} contains either only P_2 or only P_5

X_{13} effected $(P_1 + P_2)^2$

X_{14} effected $(P_1 + P_2 + P_3)^2 = (P_4 + P_5)^2$

X_{24} effected $(P_2 + P_3)^2$

X_{25} effected $(P_2 + P_3 + P_4)^2$

$\boxed{X_{35} \text{ uneffected } (P_3 + P_4)^2 = (P_1 + P_2 + P_5)^2}$

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note generally: if P_2 and P_n shifted:

$\boxed{X_{1,i+1}}$ always only contains P_2

$\boxed{X_{2,i+1}}$ always only contains P_2

$X_{3,i+1}$ contains no P_2 and no P_n

A_5 $\quad X_{13} \quad X_{14} \quad X_{24} \quad X_{25} \quad X_{35}$

zero: $\quad s_{13} = s_{14} = 0$

$X_{13} = (P_1 + P_2)^2 = s_{12}$ not effected

$X_{14} = (P_1 + P_2 + P_3)^2 = \boxed{s_{123}} \xrightarrow{\text{effected}} = \boxed{s_{43}}$

$= \cancel{(P_1 + P_3)^2} + 2P_2 \cdot (P_1 + P_3) = \boxed{s_{12} + s_{23}}$

$\searrow s_{13} = 0$

$X_{24} = (P_2 + P_3)^2 = s_{23}$ not effected

$= (\underbrace{P_4 + P_5 + P_1}) = \underbrace{(P_1 + P_4)^2}_{=0} + 2P_5(P_1 + P_4) = s_{15} + s_{45}$

$X_{35} = (P_3 + P_4)^2 = s_{34}$

$= (P_1 + P_2 + P_5)$

$s_{13} = s_{14} = s_{15} = 0$

$n = 6$

$X_{36} = (P_3 + P_4 + P_5)^2 = (P_1 + P_2 + P_6)$

$X_{35} = (P_3 + P_4)^2 = (P_1 + P_2 + P_5 + P_6)$

$X^{(0)}_{i\,j+1}$ is the $X_{i\,j+1}$ after setting $s_{1i} = 0$ $\quad (i = 3, \sim n-1)$

$2 \le j \le n-2$

$$X^{(0)}_{1\,j+1} = s_{12} + \sum_{i=3}^{j} s_{2i} + s_{3\cdots j}$$

$$X^{(0)}_{2\,j+1} = \sum_{i=3}^{j} s_{2i} + s_{3\cdots j}$$

$$X^{(0)}_{2\,n} = -s_{12}$$

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after BCFW shift $\quad X_{ij} = X_{ij}(z)$ (polynomial)

$X^{(\infty)}_{ij}$ is the leading order coefficient in z

$$X^{(\infty)}_{ij} = \frac{d}{dz}\Big|_{z=0} X_{ij}(z) \quad \Big\| \quad \text{or} \quad X_{ij}(z) = X_{ij}(0) + X^{\infty}_{ij}\, z + O(z^2)$$

shift P_2 and P_n

$$X_{ij+1}(z) = S_{12}(z) + \cancel{\sum_{i=3}^{j} S_{1i}(z)} + \sum_{i=3}^{j} S_{2i}(z) + \cancel{S_{3\cdots j}(z)} = S_{12}(z) + \sum_{i=3}^{j} S_{2i}(z) + O(1)$$

$$S_{12}(z) = z\, P_1 \cdot (P_2 + q z) = z z\, P_1 \cdot q + o(1)$$

$$S_{2i}(z) = z\, P_i \cdot (P_2 + q z) = z z\, P_i \cdot q + o(1)$$

$$X_{ij+1}(z) = z z\, P_1 \cdot q + z z \sum_{i=3}^{j} P_i \cdot q + o(1)$$

$$\begin{cases} X^{(\infty)}_{i\,j+1} = \underline{2\, q \cdot P_1} + \underline{2 \sum_{i=3}^{j} q \cdot P_i} \\ X^{(\infty)}_{2\,j+1} = \underline{2 \sum_{i=3}^{j} q \cdot P_i} \\ X^{(\infty)}_{2n} = \underline{-2\, q \cdot P_1} \end{cases}$$

$j = 2, \dots, n-2$

$n-2-3+1 = n-3$ eqs

<u>Relations:</u>

$$X^{(\infty)}_{i\,j+1} - X^{(\infty)}_{2\,j+1} + X^{(\infty)}_{2n} = 0$$

$$X^{(0)}_{i\,j+1} - X^{(0)}_{2\,j+1} + X^{(0)}_{2n} = 0$$

$j = 2, \dots, n-2$

$(n-3)$ eqs

$$X_{i\,j+1}(z) - X_{2\,j+1}(z) + X_{2n}(z) = O(1)$$